

ALI NIKKHAH

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SUMMARY

Senior AI Engineer & Technical Lead with 2+ years architecting production AI systems at scale. At Digikala, led the enterprise AI intelligence layer (5M+ daily queries) as R&D Lead: designed LangGraph multi-agent architecture, hybrid RAG with KAG, and vLLM/Triton serving stack; supervised multi-tenant K8s deployment (200 pods). At Advanced Analytics, led edge CV deployment on AXIS hardware (12 sites, 100+ cameras). At Turquoise, drove BNPL credit risk modeling (AUC 0.87) and financial agentic chatbot with voice AI. Expert in: agentic systems, LLM serving, MLOps, distributed ML, technical leadership.

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript
Deep Learning & AI: PyTorch, TensorFlow, JAX, Hugging Face, LangGraph, LangChain, LlamaIndex, FAISS, HyDE, KAG
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, TensorRT, TensorFlow Lite
Data & Streaming: Spark, Trino, Airflow, Kafka, PeerDB, Elasticsearch, Parquet
MLOps & Infra: Docker, Kubernetes, Terraform, Ray, Triton, vLLM, MLflow, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led team of 5 engineers architecting enterprise AI layer (5M+ daily queries); defined R&D roadmap transitioning from semantic matching to LangGraph multi-agent pipelines with hybrid RAG (FAISS + Elasticsearch via RRF), HyDE, and KAG (**+23% recall@10**, **-41% hallucinations**).
- Designed vLLM/Triton serving stack with PagedAttention, async FastAPI, multi-tier Redis caching achieving **3.2x throughput**, **60% cost reduction**, **p99 <200ms**; established ML best practices: eval pipelines, guardrails, A/B testing framework.
- Supervised multi-tenant Kubernetes deployment (200 pods) isolating vision/language clusters (**40% GPU cost reduction**); mentored 3 engineers on agentic systems, LLM serving, MLOps.

Advanced Analytics Australia

Remote Contract

Computer Vision Tech Lead

Sep 2025 – Apr 2026

- Led edge CV deployment across **12 industrial sites**: PPE detection, handwash verification, face tracking (**<50ms inference**, **99.2% uptime**, **-35% false positives**); TensorRT/TF Lite via Triton, Kafka for **100+ cameras @ 30 FPS**; defined sensor calibration strategy.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer & Data Scientist

Sep 2025 – Mar 2026

- Drove BNPL credit risk modeling initiative: developed ensemble models (AUC 0.87, **-22% default rate**) with automated fund allocation and audit logging; presented to C-suite for \$2M+ portfolio decisions.
- Engineered LangGraph financial chatbot with Whisper/wav2vec2.0 voice transcription (**-78% tickets**, **94% accuracy**); secure function-calling for live equity queries without exposing DB credentials.
- Built time-series transformer portfolio allocation system (Sharpe +0.42); architected data lake (Trino, PeerDB CDC **<60s latency**, **5TB/day**); PySpark ETL (**60% memory reduction**).

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Established ML platform from scratch: K8s + Ray + MLflow + TensorFlow Serving; CI/CD (ArgoCD, GitHub Actions **<10min**); FastAPI inference APIs; **15+ models**, **99.9% deploy success**; trained team on MLOps practices.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024

RESEARCH HIGHLIGHTS

Univ. of British Columbia (*Remote RA, Feb 2024 – Nov 2025*) — Vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for automated DICOM-compliant ultrasound report generation; achieved SOTA BLEU & ROUGE-L. Preprint in preparation.

L3S Research Center (*Remote RA, Apr 2023 – Feb 2024*) — 3D CNNs and transformer encoders over dense optical flow for texture-free motion classification; SOTA on UCF-101 and HMDB-51.